```
Group:1 [ backend servers ]
````````
master : DNS Server + Dedicated Apache Web Server
slave   : DNS Server + Dedicated Apache Web Server

Group:2 [frontend server]
````````
proxy   : haproxy

Group:3 [ Clients ]
````````
curl haproxy
```

---

```
Apache Clustering & Load Balancer
`````````````````````````````````````````


HAProxy [ Apache Cluster & Load Balancing ]
`````````````````````````````````````````````
apache loadbalancer
-`````````````````````
Master Apache : Dedicated Web Server
        master.sniper.com [192.168.206.134]

Slave Apache : Dedicated Web Server
        slave.sniper.com [192.168.206.149]

HA Proxy:
        haproxy.sniper.com [192.168.206.150]

++

dnf install haproxy -y

vim /etc/haproxy/test.cfg
frontend http80
        bind *:80
        timeout client 60s
        stats uri /haproxy?stats   <<<<<< Optional
        mode http
        default_backend webservers

backend webservers
        timeout connect 10s
        timeout server 100s
        balance roundrobin
        mode http
        server web1 master.sniper.com:80 check
        server web2 slave.sniper.com:80 check

haproxy -f test.cfg

CHECK: http://load_balancer_public_ip/haproxy?stats

Password protecting the statistics page
`````````````````````````````````````````
listen stats
   bind *:8181
   timeout connect 10s
   timeout client 10s
   timeout server 10s
   mode http
   stats enable
```

```
    stats uri /dashboard
    stats realm Haproxy Statistics
    stats auth admin:secret

NOW CHECK: http://load_balancer_public_ip:8181/dashboard

firewall-cmd --add-service=http --permanent
firewall-cmd --add-port=8181/tcp --permanent
firewall-cmd --reload

ALGO:
Roundrobin: Each server is used in turns according to its weights. This is the smoothest and fairest
algorithm when the servers’ processing time remains equally distributed. This algorithm is dynamic,
which allows server weights to be adjusted on the fly.
Leastconn: The server with the lowest number of connections is chosen. Round-robin is performed
between servers with the same load. Using this algorithm is recommended with long sessions, such as
LDAP, SQL, TSE, etc, but it is not very well suited for short sessions such as HTTP.
First: The first server with available connection slots receives the connection. The servers are
chosen from the lowest numeric identifier to the highest, which defaults to the server’s position on
the farm. Once a server reaches its maxconn value, the next server is used.
```